



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	37.0189	Longitude	-87.0996
Zone size (km)	10	Analysis date	2026-09-25

68

FOREST INTEGRITY SCORE / 100 Degraded Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.659	Excellent
NDVI Trend 3 years (Sentinel-2)	+0.003	Stable
10-year trend (Landsat 8/9)	+0.329	Improving
40-year trend (Landsat 1984)	+0.347	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+0.87% / year	Amelioration
NDVI change 5 years	+0.069	Improving
Leaf type	Broadleaf (Deciduous/Evergreen)	Flat leaves, high photosynthesis
Developmental stage	Mature forest (60-120 years)	Well established, dense canopy
Above-ground biomass	~38.9 t/ha	Low biomass (<50 t/ha)
Canopy Height (GEDI+S2 10m)	9.3m max / 3.3m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	662,878 t CO2	CO2 sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO2 per hectare	84.4 t CO2/ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 9,943,170	Estimated value at EUR 15/tonne CO2 (voluntary carbon market)
GEDI Tree Cover	49.2%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Temperate Forest	Broadleaf/mixed, moderate climate
Sentinel-1 Radar (VV)	-9.728 dB	Dense forest (radar)
Forest cover	96.5%	Excellent
Fire pixels detected (FIRMS/year)	808 pixels	~202 events estimated
Overall trend	stable	

Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5	
Deforestation Alert	STABLE — No significant change	
Area change (3 years)	-83.1 ha	-0.68%
Annual rate (area/yr)	-27.7 ha/yr	Loss
Area change (10 years)	+11906.7 ha	+7952.77%
Annual rate 10yr (area)	+1190.7 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	18	25
NDVI trend (3 years)	15	25
Forest cover	25	25
Sentinel-1 Radar (VV)	10	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	68	100

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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